SOLUTION ARCHITECTURE

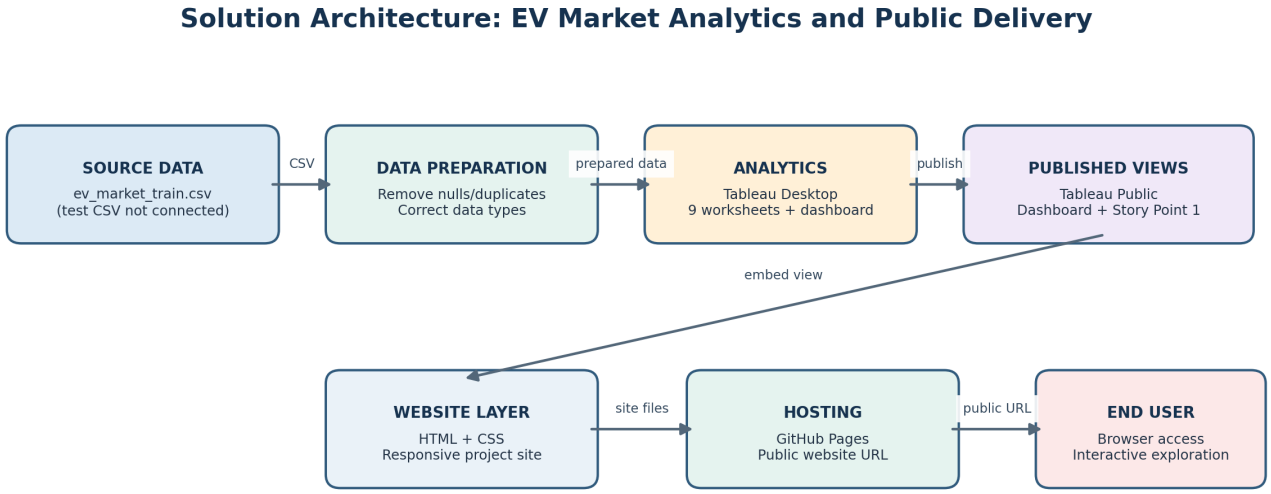
Project Design Phase-I

27 October 2026 | Team ID: LMP-EV-2026 | Yuvraj Shrivastava | EN25CS3011105

Date	27 October 2026
Team ID	LMP-EV-2026
Project	Electric Vehicle Market Performance & Insights
Project Lead / Sole Contributor	Yuvraj Shrivastava (EN25CS3011105), Medicaps University
Maximum Marks	4 Marks

1. Architecture Overview

The solution follows an analytics-to-publication architecture. A supplied CSV is prepared and analyzed in Tableau Desktop. Tableau worksheets are assembled into a dashboard and the first Tableau Story point. The workbook is published to Tableau Public, then embedded in a static HTML/CSS website hosted by GitHub Pages.



The project is a static HTML/CSS website with Tableau Public embedded for interactive visualizations.
No separate application backend, database, predictive ML model, or automated refresh pipeline is claimed.

Figure 1. Project-specific architecture. The test CSV is supplied but not connected to the Tableau workbook.

2. Architecture Components

Component Name	Description / Role in Architecture	Technology Used
Source data	Supplied EV market training dataset containing vehicle, manufacturer, technical, commercial and geographic fields.	CSV: ev_market_train.csv
Data preparation	Remove nulls and duplicate records; correct field data types. Test CSV remains unconnected.	Project data-preparation workflow; exact cleaning application not confirmed
Analytics and worksheets	Create nine analytical views, calculated aggregations and visual comparisons.	Tableau Desktop
Dashboard and story	Combine key views, global filters and chart-selection actions. Story Point 1 is completed; Story Points 2–4 are planned.	Tableau Desktop
Publication layer	Publish the workbook and interactive views for public access.	Tableau Public
Website layer	Provide project introduction, navigation and embedded Tableau visualization.	HTML and CSS
Hosting and access	Serve the static website publicly and allow browser access to the embedded Tableau view.	GitHub repository and GitHub Pages

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Data Flow, Boundaries and Ownership

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3. Data Flow

• Input: ev_market_train.csv is connected to the Tableau workbook. ev_market_test.csv is available as a supplied resource only and is not connected.

• Preparation: nulls and duplicate records are removed, and data types are corrected.

• Analysis: prepared fields are used in nine worksheets, including KPI, category sales, manufacturer share, battery performance/adoption, geographic footprint, price-range comparison, rating-related analysis and the battery matrix.

• Presentation: worksheets are arranged in the EV Market Performance & Strategy Dashboard with model year, category, battery type and manufacturing country filters.

• Publication: the Tableau workbook is published to Tableau Public and embedded in the GitHub Pages website through HTML.

• Consumption: viewers open the public website in a browser and interact with the embedded Tableau view.

4. Design Boundaries and Current Status

Area	Current status / boundary
Completed	Data preparation; nine Tableau worksheets; interactive dashboard; Tableau Public publication; GitHub Pages website; Tableau Story Point 1.
Planned	Tableau Story Points 2–4: battery chemistry/range story, pricing/customer satisfaction story, and global manufacturing footprint/strategy recommendations.
Not included / not evidenced	Separate database, REST API, Flask/Streamlit backend, machine-learning prediction model, automated refresh pipeline or dedicated cloud compute.

5. Ownership

The architecture, Tableau workbook, dashboard and website were created and published by Yuvraj Shrivastava (EN25CS3011105), Medicaps University, independently. No other contributor was involved.